

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCOC401 - DESIGN & ANALYSIS OF ALGORITHMS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. Max heap is complete binary tree.
a) True b) False
2. Which of the following provides maximum asymptotic complexity.
a) $f(n) = n^{3/2}$ b) $f(n) = n \log n$ c) $f(n) = 2^n$ d) $f(n) = n$
3. Which one of following array elements represents binary min heap ?
a) 12 10 8 25 14 17 b) 8 10 25 14 17
c) 25 17 14 12 10 8 d) 14 17 25 10 12 8
4. An Algorithm represented in the form of programming languages is _____.
a) Flowchart b) pseudo code c) program d) instruction
5. Recurrence relation can be solved using
a) Substitution method b) Master method c) Recursion tree d) all of these
6. Dijkstra's algorithm is an example of _____ approach.
a) Backtracking b) Greedy c) Dynamic programming d) Divide and Conquer
7. If for an algorithm time complexity is given by $O(1)$ then complexity of it is
a) constant b) polynomial c) exponential d) none of the mentioned
8. Matrix chain multiplication algorithm is an example of _____ approach.
a) Backtracking b) Greedy c) Dynamic programming d) Divide and Conquer
9. Merge sort uses which of the following technique to implement sorting?
a) Dynamic programming b) Greedy approach
c) Divide and conquer d) Backtracking
10. Which of the problems cannot be solved by backtracking method?
a) n-queen problem b) Subset sum problem
c) Hamiltonian circuit problem d) Travelling salesman problem
11. The worst-case time complexity of Quick sort is
a) $O(\log n)$ b) $O(n)$ c) $O(n \log n)$ d) None of these
12. All NP-complete problems are NP-hard.
a) True b) False

- Q2. Solve the following.
- A) What are different asymptotic notations ? Explain in detail. 6
- B) State and explain Master Theorem . Solve the following by applying Master Theorem.(i) $T(n) = 16T(n/4) + n$ (ii) $T(n) = 3T(n/3) + n/2$ 6
- Q3. Solve the following.
- A) How are the problems solved using divide and conquer approach ? Apply Quicksort algorithm on following array. $A = \{10, 9, 20, 40, 30, 90, 60\}$ 6
- B) Write Strassen's algorithm for multiplying 2×2 matrices . What are limitations of Strassen's algorithm. 6
- Q4. Solve Any Two of the following.
- A) Distinguish between greedy algorithm and dynamic programming approach. 6
- B) Apply Huffman Coding algorithm on following data .Obtain encoding for this data. a: 45 b:13 c:12 d:16 e:9 f:5 6
- C) Solve the following knapsack problem using Greedy approach. 6
- | Item | Weight(Kg) | Value |
|------|------------|-------|
| 1 | 3 | 10 |
| 2 | 3 | 15 |
| 3 | 2 | 10 |
| 4 | 5 | 12 |
| 5 | 1 | 8 |
- Q5. Solve Any Two of the following.
- A) Obtain two solutions to 4 queens problem using backtracking approach. 6
- B) Explain how branch and bound technique is applied to solve traveling salesman problem. 6
- C) Explain the relation between NP problem , NP hard problem , NP complete problem using Venn diagram. 6
- Q6. Solve Any Two of the following.
- A) Determine longest common subsequence between X and Y if $X = A,B,C,B,D,A,B$ and $Y = B,D,C,A,B,A$. 6
- B) Explain the steps involved in dynamic programming to solve optimization problem. 6
- C) Write and explain working of Floyd Warshall's algorithm. 6

*** End ***